

4. Setup, warm-up, and robot-to-sensor registration

Registration gives every later capture a ground-truth target pose in the sensor frame. Two things anchor it. Robot repeatability fixes relative motion. A plane-correspondence solve on the noise plate T2 fixes the camera's pose in the robot base, using the depth planes themselves. What the planes cannot observe, the sideways position of a target on the flange, comes from the as-built datum and a once-per-mount check against the left IR image (Step 4.8).

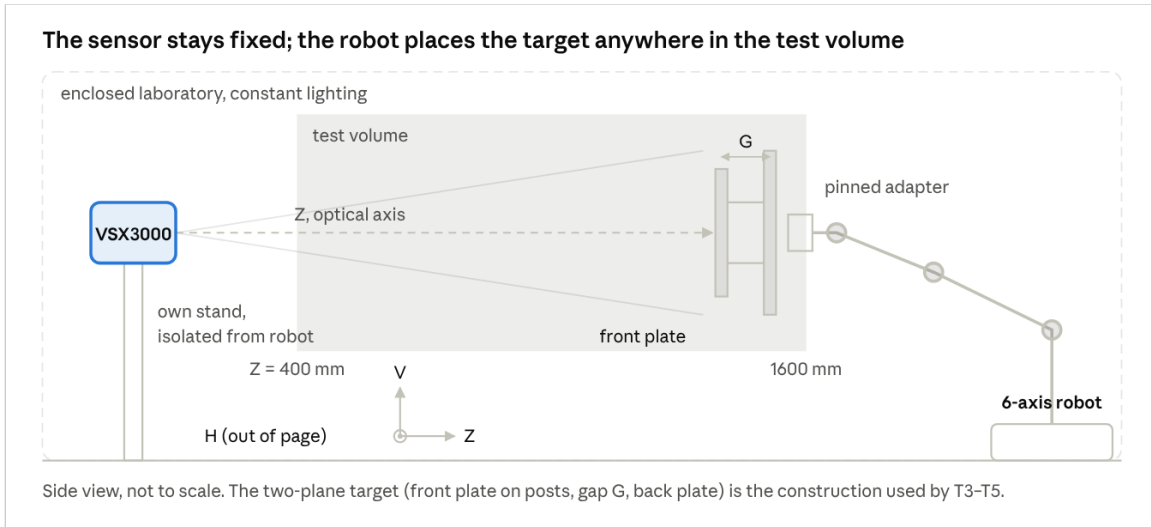


Figure 3. Experimental setup, side view

The arrangement is shown in Figure 3. After Step 4.7, the robot-base-to-sensor transform is known, so every commanded target pose is also a ground-truth pose in sensor coordinates.

1. Environment. The laboratory is enclosed and its lighting is constant; keep it unchanged for the whole session. Log the air and sensor-housing temperatures every minute (TEMPERATURE_LOG_INTERVAL_MIN).
2. Sensor configuration. Disable auto-exposure and fix exposure, gain, emitter power, and trigger mode. Record every depth-processing setting (temporal filter, spatial filter, hole filling, confidence threshold) in sensor_config.json. Characterize the configuration that production will use. If the filters can be switched off, run A and B a second time with the filters off, so the sensor's own processing can be separated from its physics; this repeat is outside the Section 9 budget. Record the SDK and firmware versions in the same file.
3. Warm-up. Power the sensor for at least 45 minutes (WARMUP_MIN_MINUTES). Then put T2 fronto-parallel at the reference station, 800 mm (Z_REFERENCE_MM), and capture 10 frames

(WARMUP_CHECK_FRAMES) every minute (WARMUP_CHECK_INTERVAL_MIN). Here σ_t is the temporal standard deviation of those warm-up frames themselves. Start testing once the mean plane Z has drifted less than a tenth of σ_t (WARMUP_DRIFT_FRACTION_OF_SIGMA) over 10 minutes (WARMUP_DRIFT_WINDOW_MIN).

Optional separate drift run. Before the session, on the same day or the day before, with the robot idle: put T2 on a fixed stand at the reference station, fronto-parallel, power the sensor from cold, and capture 30 frames (SENTINEL_FRAMES) every 2 minutes (DRIFT_RUN_CAPTURE_INTERVAL_MIN) for 480 minutes (DRIFT_RUN_DURATION_MIN), which covers the warm-up and the planned session length, while the air and sensor-housing temperatures are logged as in Step 1. The run measures the sensor's own depth drift as a function of time and of its temperature reading, with the robot's thermal growth and any mount event left out. It gives the warm-up time and the drift-versus-temperature relation that Section 10, Step 11 compares with the in-session sentinels, which remain required because they alone see the robot and the mounts on the day. The run costs no robot time and is outside the Section 9 budget; its files go in the sentinels folder with the sub-series label `drift_run`.

4. Settle and vibration check. With T2 at the near limit, 400 mm, where the depth noise is smallest and a robot vibration of a given amplitude is easiest to see, capture 100 frames (SETTLE_CHECK_FRAMES) twice: once with the servos on after a move and the settle wait, and once with the brakes engaged. If the servo-on σ_t exceeds the brakes-on σ_t by more than 10% (SETTLE_SIGMA_EXCESS_FRACTION), raise the 2 s settle time (ROBOT_SETTLE_TIME_S) or stiffen the target mount, then repeat.

5. Intrinsic and frame checks.

1. Read the left IR intrinsics, the depth-to-IR extrinsics, the baseline (SENSOR_BASELINE_MM), the depth LSB (DEPTH_LSB_MM), the projector offset, the 2-D sensor position, and the frame rate from the SDK or datasheet. Fill in the † entries in Section 2. Confirm that the sensor returns valid depth on T2 at 400 and at 1600 mm; if not, tighten the range limits (Z_MIN_MM, Z_MAX_MM) to the sensor's range and rebuild the station ladder.
2. Confirm the depth image is registered to the left IR image. With T3a in view, overlay the square's edges from the IR image on the depth discontinuities. They must agree within 0.5 px (FRAME_CHECK_PX).
3. Confirm the baseline direction and side. The occlusion band (a strip of no-reads) appears beside vertical edges only if the baseline runs along H; if it appears beside horizontal edges, swap H and V in Procedures B and E. With the right camera at +H, as Section 1 assumes, the

band lies outside the $-H$ edge of the raised square and inside the $+H$ edge of the window; if it lies on the other side, reverse the sign of H in Procedures B and E.

6. Registration capture. Mount T2. Command 30 poses (REGISTRATION_POSES) that span 400 to 1600 mm and cover the field of view, with tilts about H and about V , of both signs, up to $\pm 20^\circ$ (REGISTRATION_TILT_RANGE_DEG), so that the plate normals span all three directions; the plate must stay inside the field at every tilt. At each pose capture 10 depth frames (REGISTRATION_FRAMES) and the read-back robot pose.
7. Registration solve and acceptance. Fit the plate plane in each pose's mean depth frame (normal and distance in the camera frame). Solve the hand-eye problem from the plane correspondences: a closed-form start, then joint nonlinear least squares on the plane-normal and plane-distance residuals, giving the camera $\rightarrow$ robot-base transform and the plate's normal and offset on the flange. The plate's sideways position on the flange and its rotation about its normal are not observable from planes and are not needed. Accept if the RMS plane-distance residual is at most 0.15 mm (REGISTRATION_RESIDUAL_ACCEPT_MM), and report beside it the standard error of the camera's Z offset from the fit covariance, about 0.08 mm at the indicative noise (Section 15). If either is too large, add poses with larger tilts, up to the $\pm 20^\circ$ limit, and solve again. Save both transforms, the residual, and the standard error in registration.json.
8. Station targets and mount check. For every station in Sections 5–8, compute the robot pose that puts the target's reference point at the required (H , V , Z) in the camera frame, fronto-parallel unless stated otherwise. Re-mounting a target on the dowel-pinned adapter needs no new registration. Check each mount once at the reference station, 800 mm: fit the mounted target's front plane from the depth data and compare it with the registered pose, Z within 0.15 mm (REGISTRATION_RESIDUAL_ACCEPT_MM) and tilt within 0.05° (MOUNT_TILT_TOLERANCE_DEG); then locate one feature edge or outline in the left IR image and compare its H and V with the as-built datum offsets, within 0.5 px (FRAME_CHECK_PX). Otherwise re-seat the target or correct the datum record.